Changelog

v1.7.10 — 2026-06-28 — Round-3 audit: enable-plugins bug fix, path-traversal guards, proxy robustness
- Fixed
- Security (defense-in-depth)
- Robustness
- Verified
- Audit (round 3) — verified clean

v1.7.9 — 2026-06-28 — Hardening round 2: injection-safe alias writes, broadened secret redaction, docs accuracy, shellcheck 0
- Security
- Fixed
- Docs
- Quality
- Verified

v1.7.8 — 2026-06-28 — Secret hygiene (argv + committed-proof leaks), dead-code fix, coverage tests
- Security
- Fixed
- Added
- Verified

v1.7.7 — 2026-06-28 — Portable realpath (BSD portability hardening), set -u edge fix, regression tests
- Fixed
- Added
- Verified
- Audit findings (v1.7.6 — no code change required)

v1.7.6 — 2026-06-28 — Always-non-interactive execution, alias-file integrity, macOS/bash-3.2 portability, 4-host rollout
- Fixed
- Added
- Multi-host rollout (nezha · mistborn.local · thinker.local · amber.local)
- Verified

v1.7.5 — 2026-06-28 — Cross-provider /resume session visibility fix
- Fixed
- Root Cause Analysis
- Verified
- Tests

v1.7.4 — 2026-06-26 — Kimi provider fix + AWS IaC MCP disabled by default
- Fixed
- Changed
- Tests

v1.6.6 — 2026-06-21 — TOON integration for token-efficient prompts
- Added
- Token Savings
- Note
- Tests

v1.6.5 — 2026-06-21 — Poe proxy fix (alias file + install)
- Fixed

Verified
Tests

v1.6.4 — 2026-06-21 — Poe proxy fix for tool compatibility
Fixed
Verified
Tests

v1.6.3 — 2026-06-21 — Poe provider (382 models, 3 aliases)
Added
Verified

v1.6.2 — 2026-06-21 — Chutes provider documentation + model update
Changed
Verified

v1.6.1 — 2026-06-21 — cache_control fix + E2E tests
Fixed
Added
Verified working (all aliases tested with "Do you see our codebase?")

v1.6.0 — 2026-06-21 — Multi-alias provider system
Added
Changed
Full test suite
Usage

v1.5.1 — 2026-06-20 — Linux stat fix + nezha deployment
Fixed
Deployment
Full test suite

v1.5.0 — 2026-06-20 — Cross-alias session visibility
Added
Changed
Full test suite
How it works
Performance

v1.4.0 — 2026-06-20 — OpenCode Zen provider alias
Added
Changed
Full test suite
Honest notes

v1.3.0 — 2026-06-19 — Xiaomi MiMo provider alias
Added
Changed
Full test suite
Honest notes

v1.2.0 — 2026-06-19 — Z.AI Coding Plan provider alias
Added
Changed
Full test suite

v1.1.0 — 2026-06-16 — Distributed infrastructure + provider verification
Added
Changed

Fixed (LLMsVerifier — shipped as PR #2, applied to deployed builds)
Verified live (real "Do you see my code?" against production APIs)
Safety
v1.0.0 — 2026-06-16 — Dynamic provider-alias generator
v1.6.7 — 2026-06-21 — Poe proxy fix for all aliases
Fixed
Verified
Tests
v1.6.8 — 2026-06-21 — Poe proxy gzip fix
Fixed
Verified
Tests
v1.6.9 — 2026-06-21 — Poe proxy $ref fix for Grok-4
Fixed
Verified
Tests
v1.7.0 — 2026-06-22 — Poe proxy complete fix (all aliases verified)
Fixed
Verified (all three aliases through full Claude Code flow)
Root Cause Analysis
Tests
v1.7.1 — 2026-06-22 — Full validation + release
Fixed
Tests
Release
v1.7.2 — 2026-06-22 — Claude alias verification, full release
Added
Tests

# Changelog

All notable changes to the Claude multi-account toolkit.

## v1.10.3 — 2026-06-29 — Execution-level wrapper test coverage

### Added

- **`test_wrapper_exec.sh`** — the first hermetic test that actually *executes* the generated `cma_run` wrapper (every other suite only string-matches its emitted body, so a runtime bug — a `set -e` abort, a dropped `unset`, wrong call order — could ship past a green suite). It drives `cma_run` with a stub `CLAUDE_BIN` env-recorder plus stub `claude-session`/`claude-sync-state`, then asserts RUNTIME guarantees: provider-env isolation (a leaked `ANTHROPIC_BASE_URL/AUTH_TOKEN/MODEL` is genuinely cleared *before* claude runs), session flags reach claude on a bare launch, `sync-state pull` fires before launch and `push` after, explicit args pass through verbatim with no session-flag injection, plus a non-vacuity guard proving the stub claude really executed. Proven **RED** on a dropped `unset`, **GREEN** on the real wrapper.

### Verified

- Suite **18/18 green**; shellcheck 0.

## v1.10.2 — 2026-06-29 — Self-healing rc source lines + strict rc tests

### Fixed

- **Dangling `source "…/aliases.sh"` lines in rc files.** A transient or moved alias-file path could leave a `source` line in `~/.bashrc`/`~/.zshrc` pointing at a deleted file, so every new login shell printed `-bash: …/aliases.sh: No such file or directory`. `cma_ensure_alias_file` now **prunes** any rc `source`/`.` line whose `aliases.sh` target no longer exists (self-heal on the next install), and recognizes an existing source line across `.`/`source` and `$HOME`/~/absolute forms, so re-installs never accumulate duplicate source lines.

### Added

- **`test_rc_sourcing.sh`** (10 strict assertions) — reproduces the bug class the hermetic suite missed (it sandboxes `$HOME` and never inspected or *sourced* the rc files): prune drops dangling / keeps valid + comments + unrelated lines, ensure self-heals, **a fresh shell sources the rc with NO error** (the reported symptom), idempotent (exactly one source line after 3 calls), and cross-form dedup. Proven RED on the old behavior, GREEN on the fix.

### Verified

- Suite **17/17 green**; shellcheck 0.

# v1.10.1 — 2026-06-29 — Robust cma_run wrapper assertions

## Fixed

- **`test_claude.sh` used a fixed `grep -A30` window** to scan the `cma_run` body and silently missed the sync-state `push` marker once the body grew with the v1.10.0 apply-color calls (push slipped past line 30) — failing the suite against a v1.10.0-installed alias file even though the wrapper itself was correct. It now extracts the full function body (awk header → closing brace), robust to future growth.

## Verified

- Suite **16/16 green** against the v1.10.0 wrapper; shellcheck 0.

# v1.10.0 — 2026-06-29 — Auto-applied per-alias session color + coverage/ wiring

The per-alias session color is now **auto-applied** (it was only a hint in v1.9.x), plus self-healing for a stale `CLAUDE_BIN` and several closed test-coverage gaps.

## Added

- **Auto-applied per-alias session color.** Each bare alias launch now writes the alias's color into the session as an `agent-color` record — the exact record Claude Code's `/color` writes — via the new `claude-session apply-color`, called by `cma_run`/ `cma_run_provider` (before launch to colour a resumed session; after exit to colour a freshly-created one). Deterministic `md5(label) mod 8` over red/blue/green/yellow/purple/orange/ pink/cyan: each alias gets a stable, distinct colour, and switching the same session between aliases re-colours it. Verified **LIVE** on claude 2.1.195 — written, idempotent, persists across `--resume`. (Prompt-bar rendering must be confirmed visually: `/color` is TUI-only and `claude -p '/color x'` is a no-op, so record injection is the only non-interactive mechanism. See docs/SESSION_COLOR.md.)
- Test coverage: `test_install.sh` (executes `install.sh` in a sandbox — symlinks, alias file, idempotency), `test_verify_scripts.sh` (`model_verify.py`
    - `providers-verify.sh`), `test_session` apply-color tests (incl. the `set -e` regression), `test_coverage` B7 (`CLAUDE_BIN` resolver),

B8 (`CLAUDE_BIN` migration), B9 (apply-color wired into both wrappers). `run-proof.sh` now also runs the previously-orphaned `verify_aliases_live.sh`.

### Fixed

- **Stale `CLAUDE_BIN` self-heals.** Existing installs whose alias file pointed `CLAUDE_BIN` at a non-existent path (e.g. `~/.local/bin/claude` where npm put claude in `~/.npm-global/bin` — the amber.local case) now rewrite it to a resolved, executable claude on the next install/ensure.
- A `set -e`/`pipefail` bug where `apply-color` aborted before writing on a session that had no existing `agent-color` record.

### Verified

- Suite **16/16 ALL GREEN**; **shellcheck 0**. Color injection proven **LIVE** on real claude 2.1.195 (write / idempotent / persist-across-`--resume` / recolour-on-alias-switch).

## v1.9.2 — 2026-06-29 — Hermetic CLAUDE_BIN resolver test

### Fixed

- **`test_coverage.sh` B7 "fallback when nowhere" was not hermetic** and failed on hosts that have a real `/usr/local/bin/claude` (caught live on thinker.local): the resolver *correctly* returns the system claude there, but the test wrongly assumed "claude nowhere" was achievable under a sandboxed HOME/PATH (it can't mask absolute system paths). Dropped that one assertion; the load-bearing discovery cases (explicit `CLAUDE_BIN`, `~/.npm-global/bin` discovery) stay covered. Runtime behavior unchanged.

### Verified

- Suite **14/14 green on all five hosts** (this host, mistborn, thinker, amber, nezha); shellcheck 0.

# v1.9.1 — 2026-06-29 — CLAUDE_BIN resolves across per-host install locations

A patch found during the live multi-host rollout of v1.9.0.

## Fixed

- **Alias launches failed where claude was installed outside `~/.local/bin`.** `npm i -g @anthropic-ai/claude-code` lands in different prefixes per host (`~/.npm-global/bin`, Homebrew, `~/.local/bin`); the toolkit's fixed `CLAUDE_BIN` default mis-pointed on those hosts, so every `claudeN`/provider launch failed "No such file or directory" (amber.local needed a manual symlink to work). `cma_resolve_claude_bin` now prefers an explicit `CLAUDE_BIN`, then `$PATH`, then the known locations (`~/.local/bin`, `~/.npm-global/bin`, `/opt/homebrew/bin`, `/usr/local/bin`), with a `~/.local/bin` fallback.

## Verified

- Suite **14/14 green**; **shellcheck 0**. `test_coverage.sh` B7 covers explicit / npm-global / fallback resolution. v1.9.0's auto-session naming confirmed installed + **live-validated** (create-named + legacy-rename) on all five hosts: this host, mistborn, thinker, amber, nezha.

# v1.9.0 — 2026-06-29 — Per-project auto-sessions that actually work live + zero-coverage tests

A minor release that makes the v1.8.0 "auto session-per-project" feature do what it promised. As shipped, opening any alias gave an **unnamed** session; three root causes — all reproduced and fixed against the real `claude 2.1.195` binary, then proven **LIVE end-to-end** — are corrected here, plus test coverage for two zero-coverage utilities and a documentation refresh.

## Fixed

- **Per-project auto-session naming never actually named the session — now proven LIVE.** Three independent root causes:
  - **Legacy/unnamed sessions were never renamed.** The launcher only `--resume`'d an existing session and never passed `--name`, so a session created by an older wrapper or

by plain `claude` stayed unnamed forever. Fix: always pass `--name` on resume too. Proven live — `claude --resume <id> --name <x>` renames a previously-unnamed session (custom-title `<NONE>` → `<x>`), contradicting the docs but confirmed empirically.
  - **The session-existence check used a run-collapsing slug.** It collapsed runs of non-alnum to one `-` (`s/[^A-Za-z0-9]+/-/g`), but claude slugs **per char**, so paths with consecutive non-alnum segments (hidden dirs, `/tmp/.private`, `__pycache__`) false-negatived the lookup and **re-created** instead of resuming. Fix: per-char slug (`s/[^A-Za-z0-9]/-/g`), matching the real on-disk dir names.
  - **`cma_run` self-heal regenerated only on a missing `unset ANTHROPIC_` marker.** Wrappers predating auto-session carried that marker but not the `claude-session` one, so they never regained the integration. Fix: regenerate when **either** marker is missing.

## Added

- **`docs/SESSION_COLOR.md`** — resolves the previously dangling reference, documenting per-project auto-session naming and the honest `/color` limitation: in `claude 2.1.195`, `/color` is **TUI-only** (no CLI flag, no `settings.json` key, no env var — verified against the binary and the docs), so the toolkit can only print a deterministic per-alias hint, never auto-apply it.
- **`test_toon.sh` (9 assertions) and `test_bootstrap.sh` (39 assertions)** — both utilities previously had **zero** coverage. toon (hermetic, SKIP-if-no-node): `toon.mjs` encode/decode round-trip, the `toon_encode.py` python→mjs chain, and non-zero exit on invalid JSON. bootstrap (hermetic): `claude-bootstrap --count 2 --yes` in a sandbox `$HOME` asserting account dirs, shared symlinks, private-file isolation, alias lines, and the documented refuse-to-clobber re-run behavior.
- **`test_coverage.sh` B6** asserts the emitted `cma_run` / `cma_run_provider` bodies actually carry the auto-session integration (bare-launch guard, `claude-session flags`, `eval set --` apply, color hint) — the session script's own unit tests can't see the wrapper — plus a **self-heal regression**: a stale `cma_run` missing the `claude-session` marker is regenerated (exactly one `cma_run()`, provider-env isolation retained, aliases preserved). `test_session.sh` updated for `--name`-on-resume and a per-char-slug regression (a `/.cfg/` path must resume, not re-create).
- **Docs refresh.** README rewritten as a project landing page; `scripts/README` refreshed with the full current script inventory; the long-form guide gained "Per-project auto-session & per-alias color" and "TOON utility" sections; the `/color` notes pinned to the verified `claude 2.1.195` (superseding older `2.1.178` references).

### Verified

- Full suite **14/14 ALL GREEN**; **shellcheck 0**. The session fix proven **LIVE end-to-end** on the real `claude 2.1.195` binary — fresh create, resume, and legacy-unnamed rename all confirmed. New tests are non-vacuous (concrete expected values / negative controls). Installed live on this host and validated.

## v1.8.1 — 2026-06-29 — Merge-engine correctness + portability hardening

A patch release: an adversarial correctness audit of `claude-unify`'s merge engine plus a BSD/GNU portability pass over the test + proof tooling. All fixes/hardening — **no new features**. Housekeeping: a divergent mirror lineage that re-created `v1.7.11` (`1e975e5`) was merged back into `main` resolved to **OURS** (local already carries v1.7.11 → v1.8.0 and later fixes that supersede it), leaving a tree byte-identical to HEAD so all four mirrors converge on one lineage; the `containers` submodule was fast-forwarded to latest `main` (`71d3256` → `67ed35a`).

### Fixed

- **`history.jsonl` merge fused records across a source missing its trailing newline.** `merge_history_jsonl` `cat`'d sources into a temp first, gluing one file's last line onto the next file's first line → two entries collapsed into one invalid-JSON line. Fix: feed files straight to `awk` (fresh record per file). Regression **R1** (RED before, GREEN after).
- **`enabledPlugins` union dropped "any true".** The `jq` used `+/*` (rightmost-wins), so a plugin enabled in an earlier account but `false` in the lexically-last account ended up disabled for everyone — contradicting the documented "any true survives" guarantee. Fix: OR-of-true reduce over every account. Regression **R2**.
- **A single malformed `settings.json` aborted the whole unify — and naive guarding then risked silent config loss.** The multi-file `jq -s` ran unguarded under `set -e` (settings is item 15 of 16), halting mid-run. Merely skipping the merge was worse: `link_to_shared` still replaced each valid account's real `settings.json` with a symlink to a never-written target (a dangling link → silent loss, exit 0). Final fix (hardened after adversarial review): validate each file with `jq empty` and merge only the valid ones (a malformed sibling is excluded, not fatal), and `link_to_shared` refuses to create a link when the shared target is absent. Regression **R3** (asserts the valid account's settings stay readable, not just that unify exits 0).

- **Directory-merge conflicts were resolved by lexical account name, not recency.** `merge_dir_into_shared`'s second pass overlaid only `ACCOUNTS[-1]` (alphabetically-last) while claiming to bias toward the "most recently active" account, so a stale account sorting last could clobber fresher `memory/*.md`. Fix: overlay every account with `rsync -au` so the newest-mtime file wins each conflict, independent of name/order. Regression **R4**.
- **Rollback left dangling symlinks.** Unify symlinks every shared item into each account; for an item an account never had there is no `.preunify` backup, so rollback's restore loop never visited it and the symlink dangled once `SHARED_DIR` moved aside. Fix: after restoring backups, remove any leftover symlink whose target points into `SHARED_DIR` (skipping the shared store itself). Regression **R5**.
- **Rollback restored a non-deterministic backup.** `find -print0` was unsorted, so when a path had several `.preunify.*` backups an arbitrary one was restored. Fix: `sort -z` (timestamps are `YYYYMMDDHHMMSS` = lexical-chronological) so the oldest — the true pre-unify original — wins. Regression **R6**.
- **`test_unify.sh` B2 was a vacuous PASS.** It called `cma_realpath` (a `lib.sh` function) without sourcing `lib.sh`, so the call errored to empty and the assertion compared `"" == ""` — the symlink target was never verified. Fix: source `lib.sh` + `set +e` (matching every sibling test that uses lib functions directly). Now prints the real resolved `SHARED_DIR/plugins/cache` path.
- **Portability: 3 GNU-only constructs broke the test/proof tooling on macOS** (the shipped runtime toolkit was already clean). `readlink -f` (no `-f` on BSD) in `assert_symlink_to`/`test_unify.sh` returned empty → spurious symlink pass/fail, fixed with a self-contained `_assert_realpath` in `assert.sh` + `cma_realpath` in `test_unify.sh`; `sed -E 's/\x1b…//'` (`\xNN` is GNU-sed-only) in `run-proof.sh`/`verify_opencode_live.sh` left ANSI in, skewing `grep -c` counts, fixed by building the ESC byte via `printf '\033'`; unguarded `timeout` (GNU coreutils) in `verify_opencode_live.sh`, fixed by resolving `timeout`/`gtimeout` once and degrading if absent.
- **De-vendored `node_modules`.** `node_modules/@toon-format/toon` was committed by an accidental "Auto-commit" yet load-bearing (`scripts/toon.mjs` imports the bare specifier; `toon_encode.py` shells out to it) with nothing ever running `npm install`. Removed from the tree (`git rm --cached` + gitignore `/node_modules/`). Proven by fresh-clone simulation: `ERR_MODULE_NOT_FOUND` before, encodes correctly after.
- **`mktemp` portability.** Standardized every bare `mktemp [-d]` to the templated `mktemp [-d] "${TMPDIR:-/tmp}/cma.XXXXXX"` form CLAUDE.md prescribes (BSD `mktemp` requires a template; only GNU tolerates a bare call) across `lib.sh`, `claude-unify/providers/opencode-sync/session/install`, and the test harness (the sandbox keeps its `cma-test.` prefix for the cleanup safety check).

### Added

- **B5 token-limit guard coverage** (`test_coverage.sh`). The v1.8.0 `context_limit/max_output` path (`cma_provider_write_env` → `CMA_PROVIDER_CONTEXT_LIMIT/CMA_PROVIDER_MAX_OUTPUT` → `cma_run_provider` exporting `CLAUDE_CODE_MAX_OUTPUT_TOKENS`) shipped with **zero** tests — the only v1.8.0 fix lacking one. 4 cases / 6 concrete-value assertions: round-trip (`262144/32768`), `null`→empty normalization, 7-arg back-compat, and the emitted wrapper carrying the export.
- **`npm install` step in `install.sh`** (soft — warns, never hard-fails, when `npm` is absent; core unify/add-account needs no Node), so a fresh clone gets `@toon-format/toon` without a vendored tree. `curl-install.sh` inherits it via delegation.
- **+16 regression assertions** — 6 in `test_coverage.sh` (B5) and 10 in `test_unify.sh` (R1–R6 above), each written RED-before / GREEN-after.
- **Documented two deliberate merge/sync trade-offs** in-code so they are explicit rather than silent: `cma_merge_claude_json` replaces (not element-unions) array values; `claude-sync-state` pull/push is last-writer-wins (no portable mutex is worth its stale-lock failure modes for a per-launch hook).

### Verified

- Full suite **12/12 ALL GREEN**; **shellcheck 0**; all `.py` compile under `python3 -W error`. Each bugfix proven **RED-before / GREEN-after**; the de-vendor proven via fresh-clone simulation (`node scripts/toon.mjs` + `toon_encode.py`); ESC-strip verified functionally; the post-merge tree confirmed byte-identical to HEAD. Installed live on this host and validated against all existing aliases (3 native + 44 provider) + `claude-list-accounts`.

## v1.8.0 — 2026-06-29 — Alias isolation + token-limit guard + per-project auto-sessions

A systematic-debugging pass fixing three reported issues plus a new session-per-project feature. Every root cause was reproduced and the fix proven with physical evidence before shipping.

### Fixed

- **CRITICAL — aliases cross-contaminated API endpoints across sessions.** `cma_run_provider` exports `ANTHROPIC_BASE_URL/ AUTH_TOKEN/MODEL/ SMALL_FAST_MODEL` into the interactive shell, and native `cma_run` did **not** clear them — so running a provider alias (e.g. `xiaomi`) and then a native alias (`claude1`) in the same shell made the native one inherit the provider's endpoint (`api.xiaomimimo.com`). `cma_run` now `unset`s those four vars before launch. Proven live: after a leaked xiaomi env, native launch shows `ANTHROPIC_BASE_URL=<unset>`. Existing installs auto-regenerate the wrapper (migration keyed on the new `unset ANTHROPIC_` marker).

- **Token-limit 400 ("exceeded model token limit: 262144").** The models.dev catalog's per-model `limit.context` / `limit.output` were read for ranking but never emitted, so Claude Code overshot a provider's real context window. `providers_resolve.py` now emits `context_limit` + `max_output`; `providers_generate.py` and `cma_provider_write_env` write `CMA_PROVIDER_CONTEXT_LIMIT` / `CMA_PROVIDER_MAX_OUTPUT` into each `.env`; and `cma_run_provider` exports `CLAUDE_CODE_MAX_OUTPUT_TOKENS` from it. Proven: kimi-for-coding now resolves `context_limit=262144 max_output=32768` from the live catalog.
- **"workspace has not been trusted" warning on launch.** Confirmed NOT a merge bug (trust propagates across accounts correctly); the warned project was simply never trusted under any account. The launch wrapper now writes `projects[<root>].hasTrustDialogAccepted=true` for the launching project via the new `claude-session` helper.

## Added

- **Auto session-per-project (`claude-session.sh`).** Every bare alias launch (native or provider) now resumes — or, the first time, creates — one long-lived Claude session per project root: stable `--session-id` (UUID derived from the git-root path), `--name` set to the root dir basename in lowercase snake_case (`Android 15` → `android_15`). Explicit args/flags are always respected verbatim. Verified against the real `claude` CLI: `--session-id` creates, `--resume` resumes.
- **Per-alias color hint.** A deterministic alias→color mapping over Claude Code's real palette (`red blue green yellow purple orange pink cyan`); printed as a `/color <x>` tip on launch. (Investigated thoroughly: `/color` is a TUI-only command with no CLI flag / settings key / writable persistence, so it cannot be auto-applied — the toolkit suggests it rather than faking it.)
- **`test_session.sh`** — 27 hermetic assertions for name/id/color/flags/trust/ git-root behavior. **`run-all.sh` is now 12 files / 60 assertions, ALL GREEN.**

## Verified

- Full suite **12/12 ALL GREEN**; **shellcheck 0**; all `.py` compile under `python3 -W error`. All four items proven end-to-end against the live catalog and the emitted alias file.

# v1.7.12 — 2026-06-28 — One-line curl installer

## Added

- **`curl-install.sh`** — one-line bootstrap installer:

  ```
  curl -fsSL https://raw.githubusercontent.com/vasic-digital/
          claude-toolkit/main/scripts/curl-install.sh | bash
  ```

  Detects platform (Linux/macOS) and shell, auto-installs missing hard dependencies (jq, rsync, awk) via the system package manager (apt/dnf/apk/pacman/brew), clones (or pulls if already present) the repo with all submodules recursively to `~/claude-toolkit`, runs `install.sh`, and prints next-steps. Idempotent; re-runnable. Install dir overridable via `CLAUDE_TOOLKIT_DIR` env var.
- **README.md** — curl one-liner added at the top of the Install section.
- **`test_curl_install.sh`** — 22 hermetic tests covering syntax, permissions, URL correctness, submodule cloning, idempotency, platform detection, dependency checks, error handling, and next-steps output.

## Verified

- `bash -n` + `shellcheck 0` on `curl-install.sh` and `test_curl_install.sh`.
- `run-all.sh` **11/11 ALL GREEN** (was 10; +`test_curl_install.sh`).

# v1.7.11 — 2026-06-28 — Round-4: coverage-gap regression tests, toon recursion guard, arg validation

Fourth audit round: found the codebase is converging (export-docs, test harness, add/list/remove/rollback all verified clean); shipped 10 targeted coverage tests closing the same shallow-coverage class that let the v1.7.10 enable-plugins bug ship; plus 2 LOW code findings.

## Fixed

- **`toon_encode.py fallback_encode`**: unbounded recursion on deeply nested JSON. A crafted input with >1000 levels would blow Python's default stack and exit with an unhandled traceback

instead of encoding. Added a `_depth` guard; beyond 64 levels emits compact JSON as a safe fallback.
- **`toon.mjs` encode-file / decode-file**: running `toon.mjs encode-file` with no argument produced a confusing `TypeError` from Node's `fs.readFileSync`. Now prints `Error: encode-file requires a filename argument` + exit 1.

## Added — coverage-gap regression tests (the same class that let enable-plugins

ship)

- **B9 (HIGH) — `cma_ensure_alias_file` migration path** (`test_coverage.sh`): builds a realistic old `cma_run_provider()` body lacking `claude-sync-state`, calls `cma_ensure_alias_file`, asserts the body is migrated, the following `alias claude1=` survives, and `cma_run_provider()` appears exactly once.
- **B3 (HIGH) — `_cma_q` bash quoting in `cma_provider_write_env`** (`test_coverage.sh`): sources a `.env` with a model name containing a literal single quote and asserts it round-trips intact; also asserts an injection payload does NOT execute on source (mirrors the already-tested Python `q()`).
- **B1 (HIGH) — `absorb_default_plugins`** (`test_unify.sh`): creates a real plugin file under `$HOME/.claude/plugins/cache/` before unify; asserts it lands in `$SHARED_DIR/plugins/cache/`.
- **B2 (HIGH) — `link_default_plugin_subdirs`** (`test_unify.sh`): asserts `$DEFAULT_DIR/plugins/cache` becomes a symlink into `$SHARED_DIR/plugins/cache` after unify, and that re-running unify doesn't create a second backup.
- **B4 (MEDIUM-HIGH) — `sync_claude_md` seed branches** (`test_unify.sh`): branch (b) seeds `$DEFAULT_DIR/CLAUDE.md` and asserts it wins; branch (c) removes it and gives an account a `CLAUDE.md`, asserts that one wins.

## Verified

- `run-all.sh` **10/10 ALL GREEN** (coverage now 39+10=49 assertions; unify now 43+7=50); **shellcheck 0**; all `.py` compile under `python3 -W error`; `node --check toon.mjs` clean; toon_encode 500-level-nest no longer crashes; toon.mjs missing-arg gives clean error + exit 1.

# v1.7.10 — 2026-06-28 — Round-3 audit: enable-plugins bug fix, path-traversal guards, proxy robustness

Third audit round (deep dive on the less-covered surface: opencode_sync, claude-unify merge, the poe proxy, bootstrap). Fixes verified centrally.

## Fixed

- **`cma_enable_plugins` silently enabled NO plugins when given 3 or more** (`lib.sh`). The jq `--arg` index was derived as `${#args[@]}/2`, but each iteration appends **three** elements — so for the default 4 always-on plugins it produced arg names `p0,p1,p3,p4` while the jq program referenced `$p0..$p3`; `$p2` was undefined, jq failed, `2>/dev/null` swallowed the error, and `enabledPlugins` was left empty. Replaced the derived index with a dedicated counter. Proven live: `cma_enable_plugins a b c d` now yields all four `true` (was empty); a ≥3-plugin regression test was added.

## Security (defense-in-depth)

- **Path traversal via unvalidated provider id** in `claude-providers.sh` `cmd_show` / `cmd_remove`: `$id` was interpolated into `<dir>/$id.env` and then `cat`/`rm -f`'d without validation. Now rejected unless it matches `[A-Za-z0-9._-]` (blocks `../`), matching `cma_provider_write_alias`.
- **`opencode_sync.py` `${CLAUDE_PLUGIN_ROOT}` path traversal**: a malicious installed plugin could set an arg like `${CLAUDE_PLUGIN_ROOT}/../../../tmp/evil.js` and `--enable-all-local` would have OpenCode exec the traversed path. Expansion now lexically contains the result to the plugin dir; an escaping value is left unexpanded (fails safe).
- **`cma_write_alias` now rejects whitespace in the config dir** (`lib.sh`): an unquoted space silently word-split the alias into a bogus command — now a clear error instead of a broken alias.

## Robustness

- **`poe_proxy.py`**: `resolve_refs` gained a recursion-depth guard (a circular `$ref` previously crashed the request handler with `RecursionError`); the success-path `gzip.decompress` is now guarded like the error path, so a corrupt gzip body no longer propagates.

### Verified

- `run-all.sh` **10/10 ALL GREEN** (coverage now 32 assertions incl. the enable-plugins + injection regressions); **shellcheck 0**; all `.py` compile under `python3 -W error`.
- `cma_enable_plugins` fix proven live (4 plugins → all `true`); opencode containment + id validation proven with PoCs. The model-verification / alias- write path is unchanged from v1.7.9's live-proven 137 models / 32 aliases.

### Audit (round 3) — verified clean

`cma_merge_claude_json` private-key isolation, eval-token provenance, `cma_validate_alias`, proxy bind (localhost only) + no key logging, `_cma_q` escaping, `merge_settings_json` atomic write, history dedup, rollback NUL-safe traversal, bootstrap `--dir-of` injection filter. (`opencode_sync --enable-all` intentionally bypasses the needs-secret guard — operator opt-in, documented.)

## v1.7.9 — 2026-06-28 — Hardening round 2: injection-safe alias writes, broadened secret redaction, docs accuracy, shellcheck 0

A second multi-agent audit + hardening pass on top of v1.7.8 (adversarial security audit + doc-accuracy audit + lint sweep, fixes verified centrally).

### Security

- **Provider id / config dir can no longer inject shell via the alias file** (`lib.sh cma_provider_write_alias` / `cma_write_alias`). Both interpolate values into `alias name="…"` lines that the shell **re-parses on invocation**, and jq `@tsv` does not escape `"`. They now reject shell metacharacters (provider id restricted to `[A-Za-z0-9._-]`; config dir rejects `" $ \ \` ; & | < > ( )`and newline). Proven:` afoo"; touch …` payload is rejected, no command runs, the hostile alias is never written.
- **Keys-file read no longer breakable by a quote in the path** (`claude-providers.sh cmd_sync_multi`). The old `bash -c "set -a; source '$keysf'; …"` let a single quote in the keys-file path break out of the string. Replaced with an isolated subshell `( set +e; set -a +u; . "$keysf"; set +a; eval … )` — the same safe pattern `cmd_sync` already used. Proven with a `do n't/` path.
- **`.env` value quoting** (`providers_generate.py`): `q()` now POSIX single-quote-escapes embedded quotes (mirrors `lib.sh _cma_q`), so a catalog value containing a quote can't inject when

`cma_run_provider` sources the `.env`. Proven: an injection payload is neutralized to a literal.
- **xtrace secret leak** (`lib.sh cma_run_provider`): the indirect key read is now wrapped in `set +x`/restore so an active `set -x` in the user's shell can't echo the key to the terminal or a redirected log.
- **Broadened secret redaction + guard**: `cma_redact_secrets()` (`verify_opencode_live.sh`) and the committed-proof scan guard (`test_lib.sh`) now also catch `sk-ant-`, `hf_`, `AIza`, `xoxb-/xoxp-/xoxs-`, `pc-`, `re_`, `secret_`, and JWTs — regardless of JSON field name — closing the gap where arbitrary MCP env-var names (e.g. `NOTION_API_KEY`) slipped through the original six-name allowlist.

## Fixed

- **`install.sh` used `readlink -f`** (absent on BSD/macOS) for its symlink up-to-date check — missed by the v1.7.7 sweep. Now uses `cma_realpath`; the `test_lib.sh` guard scans `install.sh` too.
- **`verify_aliases_live.sh` hardcoded one developer's account dirs**, producing false FAILs on every other host. Now discovers accounts dynamically and skips dirs that don't exist.
- Dead code / cruft: `providers_generate.py` (unused import, dead vars, lambda→def, a no-op `provider_id + ('' if … else '')`); `model_verify.py` (unused `import hashlib`); `model_verify.py` docstring `--key` → `CMA_PROBE_KEY`.

## Docs

- Long-form doc + `README`s + `CLAUDE.md` corrected against the code: macOS rc-file caveat (`~/.zshrc` only), the test table now lists all 10 suites, the full installed-command list (+`claude-providers`/`claude-sync-state`/ `claude-bootstrap`), repo-relative paths (was `~/Documents/scripts/`), a new `claude-bootstrap` section, the `CMA_PROBE_KEY` security model in §11, and a refreshed date stamp.

## Quality

- **shellcheck: 93 → 0** across all scripts. Added `.shellcheckrc` (`external-sources=true`) which resolves the sourced-file warnings properly; fixed the `$?`-after-condition (SC2319) test idioms, SC2015/SC1090/SC1003; the one remaining reserved no-op flag carries a justified inline disable.

## Verified

- `scripts/tests/run-all.sh` **10/10 ALL GREEN**; **shellcheck 0**; every `.py` compiles under `python3 -W error`.

- Injection PoCs (provider id, config dir, `.env` value, keys-file path) all proven neutralized; broadened redaction proven against `AIza`/`hf_`/JWT/etc.
- Live `sync --multi`: **137 models verified, 32 aliases** across 8 providers (opencode 4, poe 33, chutes 7, huggingface 6, nvidia 30, openrouter 14, siliconflow 38, xiaomi 5), zero `CMA_PROBE_KEY`/unbound errors — identical to the v1.7.8 baseline, so the new key-read path is non-regressive end-to-end.
- 4-host byte-parity + 10/10 suite re-verified after deploy.

# v1.7.8 — 2026-06-28 — Secret hygiene (argv + committed-proof leaks), dead-code fix, coverage tests

Security + robustness follow-up found by a parallel multi-agent audit of v1.7.7. Four independent subagents fixed disjoint file sets; integration + the full suite + live multi-model verification were run centrally.

## Security

- **API key no longer passed on `argv`** (`model_verify.py` + `claude-providers.sh`). `cmd_sync_multi` invoked `model_verify.py --key "$token"`, placing the secret verbatim in `/proc/<pid>/cmdline` and `ps aux` output — readable by any user on a multi-user host. The key now flows via the `CMA_PROBE_KEY` environment variable (set per-command, not exported); `model_verify.py` reads it from the environment and errors clearly if unset. The `--key` flag is removed entirely.
- **API key no longer passed to `curl` on `argv`** (`verify_aliases_live.sh`). Six live-probe calls used `-H "Authorization: Bearer $key"`. The header is now written to a `mktemp`'d, `chmod 600` config file consumed via `curl --config` (portable on GNU + BSD curl) and removed via an `EXIT`/`INT`/`TERM` trap.
- **Leaked secrets purged from committed proof artifacts** (committed in 24bc379, rolled into this release): the OpenCode live verifier wrote resolved `opencode debug config` / `mcp list` output — which contained a real provider key and a DB connection-string password — verbatim into the committed proof dir. The three artifacts are redacted; the generator (`verify_opencode_live.sh`) now redacts via `cma_redact_secrets()` before writing (raw dump → `.raw` temp → redacted file → `.raw` removed). **Operator follow-up still required:** rotate the leaked key and decide on a git-history scrub — the values remain in history on all four remotes.

## Fixed

- **Unreachable code** in `verify_aliases_live.sh`: `exit $failed` sat *before* the Claude-alias test function and its caller, making them dead (shellcheck SC2317). `exit $failed` moved to the final statement.
- **Fragile `$?` capture** in `test_list.sh`: `grep …; [[ $? -ne 0 ]]` then `assert_eq 0 $?` read `$?` from the wrong command. Now captures `rc=$?` immediately.
- **Unquoted glob** in `claude-sync-state.sh:67`: `"$HOME"/${ACCOUNT_PREFIX}prov-*/` → `"$HOME/${ACCOUNT_PREFIX}"prov-*/` so only the intended `*` globs.
- **`SyntaxWarning: invalid escape sequence '\ '`** in `providers_resolve.py`: the usage docstring's `\` line-continuations are now a raw string (`r"""`).

## Added

- **`test_coverage.sh`** — 11 new hermetic tests (19 assertions) covering previously-untested `lib.sh` behavior: `cma_ensure_alias_file` (fresh / idempotent / old-format migration preserving unrelated lines), `cma_can_prompt` (`CMA_NONINTERACTIVE=1` and no-tty both non-interactive), `cma_enable_plugins` (JSON shape + additive + the jq `//` falsy-vs-null upgrade), `cma_link_shared_items` (every `CMA_SHARED_ITEMS` entry becomes a symlink into `$SHARED_DIR`, idempotent), and `stats-cache.json` newest-by-mtime selection.
- **Proof regression guard** (`test_lib.sh`): scans `scripts/tests/proof` for provider-key prefixes and URL `user:password@` creds, counting suspect lines so a failure never re-echoes a secret.

## Verified

- `scripts/tests/run-all.sh` — **10/10 ALL GREEN** locally (was 9; +`test_coverage.sh`).
- Live multi-model verification (`claude-providers.sh sync --multi`, real HTTP probes with the host's real keys): **137 models verified, 32 aliases generated** across 8 providers (opencode 4, poe 33, chutes 7, huggingface 6, nvidia 30, openrouter 14, siliconflow 38, xiaomi 5). Zero `CMA_PROBE_KEY`-unset and zero `unbound variable` errors — the env-var key path works end-to-end. (Providers with 0 verified are external: dead/paid keys, HTTP 401/402/403, WAF blocks — not toolkit regressions.)
- The new proof secret-scan guard immediately earned its keep: on first cross-host run it flagged a **stale, pre-redaction proof dir on all three remote hosts** (3 files with literal secrets), which were then re-synced with the redacted artifacts.
- `model_verify.py` / `providers_resolve.py` compile clean under `python3 -W error`.

# v1.7.7 — 2026-06-28 — Portable realpath (BSD portability hardening), set -u edge fix, regression tests

Follow-up hardening release found by a parallel multi-agent audit of v1.7.6.

## Fixed

- **`readlink -f` → portable `cma_realpath`** at three sites: `claude-unify.sh` (`already_linked_to_shared` and `merge_settings_json`) and `claude-list-accounts.sh` (the link check). `readlink -f` is absent on older macOS and on other BSDs (FreeBSD/NetBSD); there the checks silently fail — making `claude-unify` re-link every shared item on each re-run (accumulating stale `.preunify.*` backups) and `claude-list-accounts` report linked accounts as "not linked". **Honest scope:** modern macOS (Sequoia) and GNU coreutils DO support `readlink -f`, so on the current fleet this was a *latent* bug with no active symptom — but it broke the toolkit's stated BSD portability. Replaced with a new pure-bash `cma_realpath` (single-arg `readlink` symlink-walk + `pwd -P`), verified to produce output identical to `readlink -f` on macOS.
- **`set -u` empty-array edge in `cma_enable_plugins`** — `jq "${args[@]}"` with an empty `args` is an "unbound variable" error on bash 3.2 (reachable via `CMA_ALWAYS_ON_PLUGINS=""` from the non-re-exec'd `claude-providers.sh`). Guarded with `${args[@]+"${args[@]}"}`.

## Added

- **`cma_realpath`** portable canonicalizer in `lib.sh`.
- **Regression tests** (`test_lib.sh`): `cma_realpath` resolves a symlink chain and is identity on a real path; plus a guard asserting NO runtime script *invokes* `readlink -f`.

## Verified

- `scripts/tests/run-all.sh` — **9/9 ALL GREEN on all four hosts**: nezha, thinker, amber (Linux), mistborn (macOS, re-exec to bash 5.3, BSD userland).
- `cma_realpath` output confirmed byte-identical to `readlink -f` on macOS.

## Audit findings (v1.7.6 — no code change required)

- Disabled providers are EXTERNAL, not toolkit bugs (toolkit correctly disabled them on failed verify): `github-models` → HTTP

401 (dead GitHub PAT), `upstage` → HTTP 403 from AWS WAF (egress-IP block).
- `api_keys.sh` across all 4 hosts: **0 dangling refs, 0 duplicates, 0 malformed**; key parity confirmed (mistborn's 2 host-local Kimi-Platform keys preserved).
- Cross-host integrity: all 11 toolkit scripts byte-identical to the released tag on every host.
- Known/deferred: published tags `v1.2.0` (gitlab) and `v1.5.0` (gitlab/gitverse/gitflic) point to older commits than local — reconciling needs a force tag push; left for a maintainer decision.

# v1.7.6 — 2026-06-28 — Always-non-interactive execution, alias-file integrity, macOS/bash-3.2 portability, 4-host rollout

## Fixed

- **Alias-file corruption from a mis-firing migration** — `cma_ensure_alias_file`'s "outdated `cma_run_provider`" migration grepped for `claude-sync-state pull`, but the emitted on-disk text is `…/claude-sync-state" pull` (a quote precedes the space), so the guard **never matched** and the migration fired on *every* alias write. Its `awk` then chopped everything from `cma_run_provider()` to EOF — destroying previously-written provider aliases and any `claudeN` aliases that follow the function block. This silently corrupted the alias file on multi-provider / multi-account hosts. Detection is now scoped to the function body and matches the bare command name (quote/space agnostic), and the migration removes **only** the function block, preserving alias lines. This was the single root cause of the failures across `test_providers.sh`, `test_claude.sh`, and `test_add_remove.sh`.
- **`set -u` abort while sourcing the keys file** — provider sync sourced `~/api_keys.sh` inside a `set -euo pipefail` subshell. A dangling reference in the user's keys file (e.g. `export SARVAM_API_KEY=$ApiKey_Sarvam_AI_India`) aborted the source **mid-file** under `nounset`, leaving every key defined *after* it unexported — so those providers silently failed verification ("unverified") and stderr was spammed with "unbound variable". Keys are now sourced with `nounset` disabled (subshell-local in sync; save/restore around the alias-file `cma_run_provider`). Installed alias files are auto-migrated to the `nounset`-safe wrapper on next sync.
- **macOS / bash-3.2 portability of the test harness** — `tests/run-all.sh` used `mapfile` (bash 4+), so the **entire suite failed to run on stock macOS**. Replaced with a portable `read` loop

and guarded empty-array expansion under `set -u`. Same fix applied to `test_lib.sh` and `tests/lib/sandbox.sh` (empty `${arr[@]}` expansions are unbound on bash 3.2). The suite now runs green on macOS bash 3.2.

## Added

- **`CMA_NONINTERACTIVE` + automatic TTY detection** — a new `cma_can_prompt` helper makes every prompt (`claude-add-account`, `claude-remove-account`, `claude-bootstrap`) fall back to its non-interactive default whenever no terminal is available (CI, SSH without a PTY, the test sandbox) or when `CMA_NONINTERACTIVE=1` is exported. Toolkit execution is now **always non-interactive off a terminal**. Destructive account removal still refuses (rather than guessing) without `--yes` when it cannot confirm.
- **Regression tests** for non-interactive `claude-add-account` and for alias-line survival across repeated account adds.
- **`test_export.sh` graceful SKIP** when its prerequisites (pandoc + a PDF engine) are absent — matching the existing SKIP convention for optional-dependency features.

## Multi-host rollout (nezha · mistborn.local · thinker.local · amber.local)

- Distributed `~/api_keys.sh` to every host via a **no-loss merge** (host-local keys preserved — e.g. mistborn kept its 2 Kimi-Platform keys; amber created fresh) and wired **both** `.bashrc` and `.zshrc` to source it on every host.
- Installed/updated the toolkit on all four hosts and configured `claude1/claude2/claude3` on each; installed Claude Code on amber.
- Ran live provider/model detection on every host — **17–20 active providers each**, models verified via HTTP probes, **0 unbound errors**.

## Verified

- `scripts/tests/run-all.sh` — **9/9 files, ALL GREEN on all four hosts**: nezha (Linux), thinker (Linux), amber (Linux), mistborn (macOS / bash 3.2).
- Cross-host: both rc files source `api_keys.sh`; `claude1/2/3` + `poe/deepseek/xiaomi` aliases present on every host.

# v1.7.5 — 2026-06-28 — Cross-provider / resume session visibility fix

## Fixed

- **Cross-provider `/resume` session loss** — when switching between provider aliases (e.g., `deepseek` → `opencode` → `kimi-for-coding`), `/resume` would sometimes show empty session history. Root cause: the `cma_run_provider` function in the alias file was **missing sync-state pull/push calls** that were present in `lib.sh`. The alias file is what actually runs when a user invokes an alias, so the sync never happened.
- **Migration for outdated alias files** — added automatic detection and regeneration of outdated `cma_run_provider` functions in `lib.sh`. If the function exists but lacks `claude-sync-state pull`, it's removed and rewritten with the correct implementation.
- **Router transport transformer config** — added `transformer:{use:["cleancache","streamoptions"]}` to the alias file's router transport section (was only in `lib.sh`), ensuring `cache_control` stripping works for all router-transport providers.

## Root Cause Analysis

The `cma_run_provider` function in `lib.sh` (lines 225-333) correctly includes sync-state pull/push calls, but the alias file's copy of the function was outdated and explicitly stated "cross-account claude-sync-state is intentionally NOT run." This meant:

1. Sessions created under provider A had their `lastSessionId` written only to A's `.claude.json`
2. When switching to provider B, B's `.claude.json` still had its own (different) `lastSessionId`
3. `/resume` read B's `lastSessionId` and couldn't find A's session

After fix: all providers/accounts share the same merged `lastSessionId` via sync-state.

## Verified

- **Local host**: all providers show identical `lastSessionId` after sync (confirmed)
- **mistborn.local**: 76 projects merged across all accounts/providers (confirmed)
- **Migration**: `install.sh` correctly detects and fixes outdated alias files on both hosts

### Tests

- Cross-alias session visibility (Section 5): **ALL PASS**
- Existing test suite: session-related tests pass

## v1.7.4 — 2026-06-26 — Kimi provider fix + AWS IaC MCP disabled by default

### Fixed

- **Kimi Code provider base URL** in `scripts/providers/overrides.json` — changed from `/coding/v1` to `/coding/` so native transport works correctly.
- **AWS IaC MCP timeout** — removed `aws-dev-toolkit/awsiac` from the default OpenCode MCP allowlist in `scripts/claude-opencode-sync.sh`. The server consistently timed out on connection and is now configured but disabled by default.

### Changed

- Regenerated `Claude_Multi_Account_Fine_Tuning.{html,pdf,docx}` from current markdown.
- Refreshed proof artifacts in `scripts/tests/proof/`.

### Tests

- Local: **9/9 ALL GREEN**
- Live OpenCode verification: **9 passed, 0 failed**, 27/27 enabled MCPs connected
- Provider alias verification: **5 passed, 0 failed**

## v1.6.6 — 2026-06-21 — TOON integration for token-efficient prompts

### Added

- **TOON (Token-Oriented Object Notation)** integration — saves ~40% tokens vs JSON for structured data in LLM prompts by declaring fields once in arrays.
- **`scripts/toon.mjs`** — Node.js TOON utility (encode/decode/demo)
- **`scripts/toon_encode.py`** — Python wrapper for TOON encoding
- **`docs/TOON_Integration.md`** — comprehensive guide on using TOON with Claude Code
- **`package.json`** — `@toon-format/toon` v2.3.0 dependency

### Token Savings

- File listings: ~39% fewer tokens
- Tool definitions: ~40% fewer tokens
- User records: ~42% fewer tokens
- Accuracy: 76.4% (vs JSON's 75.0%)

### Note

TOON formats message CONTENT for token savings. API transport remains JSON (providers require it). HTTP/3 and compression require provider-side support.

### Tests

- 8/8 ALL GREEN

## v1.6.5 — 2026-06-21 — Poe proxy fix (alias file + install)

### Fixed

- **Poe proxy not starting from alias** — proxy logic was only in `lib.sh`, not in the alias file's `cma_run_provider` function. The alias file is what actually runs when a user invokes an alias. Added proxy detection + auto-start to the alias file.
- **install.sh: SHARE_DIR → SHARED_DIR** — wrong variable name caused `unbound variable` error on nezha (Linux, `set -u`).
- **install.sh: auto-copy proxy scripts** to `~/.local/share/.../proxy/` during install.

### Verified

- All 3 Poe aliases work: `poe` ✅, `poe2` ✅, `poe3` ✅
- Deployed to both local host and nezha.local

### Tests

- 8/8 ALL GREEN

# v1.6.4 — 2026-06-21 — Poe proxy fix for tool compatibility

### Fixed

- **Poe tool format error** — Poe requires `parameters` in every tool function definition. Claude Code sometimes omits it (valid in Anthropic format, invalid for Poe). Added `poe_proxy.py` that auto-fixes tools before forwarding to Poe API.
- **Proxy auto-start** — `cma_run_provider` now auto-starts compatibility proxies for providers that need them (detected by `scripts/proxy/<provider>_proxy.py`).

### Verified

- All 3 Poe aliases work through proxy: `poe` ✅, `poe2` ✅, `poe3` ✅

### Tests

- 8/8 ALL GREEN

# v1.6.3 — 2026-06-21 — Poe provider (382 models, 3 aliases)

### Added

- **Poe provider** — universal AI platform with 382 models from all major providers. OpenAI-compatible API at `https://api.poe.com/v1`. Chat, code, image gen, video gen, TTS, STT, and more.
- **3 aliases**: `poe` (claude-sonnet-4.6 + gpt-5.4-mini), `poe2` (gpt-5.5 + deepseek-v4-pro-e), `poe3` (grok-4 + gemini-3.1-pro)
- **key-aliases**: `POE_API_KEY` + `ApiKey_Poe` → `poe`
- **Tool calling verified** on claude-sonnet-4.6, gpt-5.4-mini, deepseek-v4-pro-e, grok-4
- **382 models categorized**: 130 chat/reasoning, 16 code, 40 image gen, 17 video gen, 12 TTS, 1 STT, 166 other
- **Documentation**: full Poe section in Provider_Aliases_User_Guide.md

### Verified

- API endpoint responds correctly
- Authentication works
- Tool calling confirmed

- All 3 aliases tested through ccr with "Do you see our codebase?" — all YES

# v1.6.2 — 2026-06-21 — Chutes provider documentation + model update

## Changed

- **Chutes provider models updated** — catalog was stale. Chutes now offers 13 TEE (Trusted Execution Environment) models. Updated strong=`zai-org/GLM-5.2-TEE`, fast=`Qwen/Qwen3.6-27B-TEE`.
- **Chutes documentation** added to Provider_Aliases_User_Guide.md with full model table, TEE explanation, pay-per-use note, and setup instructions.

## Verified

- Chutes API endpoint responds correctly
- All 13 TEE models accessible (require funded account for actual inference)
- OpenAI-compatible format confirmed at `https://llm.chutes.ai/v1`

# v1.6.1 — 2026-06-21 — cache_control fix + E2E tests

## Fixed

- **`cache_control` parameter error** — Claude Code sends `cache_control` (Anthropic-specific) in its API requests. ccr forwarded this to OpenAI-compatible endpoints which reject it with HTTP 422. Fixed by adding ccr's built-in `cleancache` transformer to every provider config, which strips `cache_control` before forwarding to the provider.

## Added

- **`alias_e2e_test.py`** — end-to-end alias verification script that tests each alias by sending requests through ccr and verifying responses work without errors.

### Verified working (all aliases tested with "Do you see our codebase?")

- `opencode` (north-mini-code-free): ✅ YES
- `opencode2` (big-pickle): ✅ YES
- `opencode3` (nemotron-3-ultra-free): ✅ YES
- `deepseek` (native transport): ✅ YES
- `deepseek2` (router transport): ✅ YES
- `xiaomi` (native transport): ✅ YES
- `zai-coding-plan` (router transport): ✅ YES

# v1.6.0 — 2026-06-21 — Multi-alias provider system

## Added

- **Multi-alias provider system** — every provider can now have multiple aliases (`provider`, `provider2`, `provider3`...) exposing ALL working models, not just the top 2. Verified via live HTTP probes with anti-bluff detection.
- **`model_verify.py`** — comprehensive model verification & scoring engine. Tests every model for a provider via HTTP probes, scores on 7 dimensions (existence 25pts, tool_call 20pts, reasoning 15pts, context_window 15pts, streaming 10pts, latency 10pts, free_tier 5pts). Anti-bluff detection prevents false positives (HTTP 200 with error body, empty responses, boilerplate errors). 24h verification cache to avoid re-testing.
- **`providers_generate.py`** — multi-alias generation from verified models. Pairs models into alias groups of 2 (strong + fast), handles odd count (last model reused for both positions), single model (used for both positions). Generates env files, shell aliases, and overrides.json entries.
- **`claude-providers.sh --multi`** — new flag for `sync` that triggers the full verification + multi-alias generation pipeline. Additional flags: `--max-aliases` (default 5), `--min-score` (default 25), `--verify-concurrency` (default 5).
- **Endpoint normalization** — `/anthropic` endpoints auto-converted to `/v1` for OpenAI-compatible probing during verification.
- **Submodules updated** to helix_translate-2.3.1: LLMsVerifier (ModelVerifier, Seed, xiaomi provider), challenges (anti-bluff §11.4, chaos/stress tests), containers (deploy-stack).

## Changed

- Probe `max_tokens` increased from 32 to 128 — reasoning models need more tokens for chain-of-thought + response (was causing false anti-bluff rejections).

- `User-Agent` header added to HTTP probes (some APIs require it).

### Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures.

### Usage

```
# Standard sync (2 models per provider, as before)
claude-providers sync

# Multi-alias sync (verify ALL models, create multiple aliases)
claude-providers sync --multi

# With options
claude-providers sync --multi --max-aliases 10 --min-score 20
```

## v1.5.1 — 2026-06-20 — Linux stat fix + nezha deployment

### Fixed

- **`stat -f %m` on Linux** — the mtime cache check in `claude-providers.sh` used `stat -f %m || stat -c %Y` as an `||` chain. On Linux, `stat -f` succeeds (returning filesystem info, not mtime), so both outputs merged into garbage (`"File: ...1781634386"`), causing `File: unbound variable` under `set -u`. Fixed with `case "$(uname -s)"` to pick the correct flag per platform.

### Deployment

- **nezha.local** (Linux x86_64) deployed and verified: 19 providers activated, 100/100 provider tests pass, 5/5 live verifier pass, cross-alias sync confirmed. Evidence in `scripts/tests/proof/90-nezha-deployment.txt`.

### Full test suite

- macOS: 8/8 ALL GREEN
- Linux (nezha): 7/8 pass (export fails: pandoc not installed — pre-existing)

# v1.5.0 — 2026-06-20 — Cross-alias session visibility

## Added

- **Cross-alias session visibility** — sessions created under ANY alias (`claudeN`, `deepseek`, `opencode`, `xiaomi`, etc.) are now visible from every other alias via `/resume`. Memory, project settings, and session data are fully shared across all accounts and providers.
- **`claude-sync-state.sh` extended** — now discovers provider dirs (`~/.claude-prov-*`) alongside account dirs for its `.claude.json` merge. Provider sessions participate in the same lightweight jq merge that keeps account sessions in sync.
- **`cma_run_provider` sync-state hooks** — the provider wrapper now calls `claude-sync-state pull` before launch and `claude-sync-state push` after exit, matching the `cma_run` pattern. Previously provider sessions were intentionally excluded from sync; now they participate fully.
- **Sandbox test coverage**: 10 new assertions proving cross-alias merge (sessions from account→provider, provider→account, account→account all visible after sync). Providers test 90 → 100 assertions.
- **Live verification**: `lastSessionId` for a real project confirmed identical across all dirs (3 accounts + 1 provider). 61 projects merged in every `.claude.json`. Evidence in `scripts/tests/proof/80-cross-alias-sessions.txt`.

## Changed

- `scripts/claude-sync-state.sh` — provider dirs included in merge targets
- `scripts/lib.sh` — `cma_run_provider` wrapper updated with sync-state pull/push
- Alias file `aliases.sh` — updated `cma_run_provider` function (re-installed)

## Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 10 new assertions for cross-alias session visibility.

## How it works

1. `claude-sync-state pull` merges every account's + provider's `.claude.json` into the launching dir before Claude Code starts (including `lastSessionId`, `allowedTools`, MCP config, etc.).

2. Claude Code launches with the merged state — `/resume` sees all sessions.
3. `claude-sync-state push` merges the post-session `.claude.json` back out after exit, so the next alias to launch picks up the new session.
4. The `sessions/` directory was already shared via symlink — this release ensures `.claude.json` project settings are also merged.

### Performance

- Adds ~1-2 seconds overhead per provider launch (jq merge of `.claude.json` across all dirs). Same overhead that `claudeN` aliases already have.

## v1.4.0 — 2026-06-20 — OpenCode Zen provider alias

### Added

- **`opencode` provider alias** — OpenCode Zen curated AI gateway with **21 free models** (all $0 cost, all support tool calling + reasoning) and 49 paid models. The alias uses **router transport** (ccr) targeting the OpenAI-compatible endpoint `https://opencode.ai/zen/v1/chat/completions`.
- **Model overrides**: strong = `big-pickle` (free stealth model, 200K context, reasoning + tool_call), fast = `deepseek-v4-flash-free` (free, 200K context, reasoning + tool_call). Pinning is deliberate — auto-selection would pick `nemotron-3-ultra-free` (1M ctx) as strong and `trinity-large-preview-free` (131K, no reasoning) as fast, both suboptimal for coding workloads.
- **key-aliases.json mappings**: `ZEN_API_KEY` → `opencode` and `ApiKey_Opencode_Zen` → `opencode` (both key vars present in the user's keys file).
- **overrides.json pin**: `strong_model=big-pickle`, `fast_model=deepseek-v4-flash-free` (no transport/base_url override needed — catalog values are correct).
- **Sandbox test coverage**: resolver tests (key-alias mapping for both key vars, router transport from `@ai-sdk/openai-compatible` npm, zen/v1 base_url from catalog, model override beats auto-selection, stale-model-never-selected guards) + sync e2e tests (env file, alias, config-dir + plugins symlink, account-detection exclusion, idempotency, no-secret-leak). Providers test 69 → 90 assertions.
- **Live endpoint verification**: `GET /v1/models` HTTP 200; `POST /v1/chat/completions` round trip HTTP 200 with correct text for `big-pickle` (stealth, cost=$0, reasoning_content present) and `deepseek-v4-flash-free` (cost=$0); additional free models (`mimo-v2.5-free`, `nemotron-3-ultra-free`, `north-mini-code-free`) all HTTP

200 with cost=$0. Evidence in `scripts/tests/proof/70-zen-live.txt` (secret-free).
- **Docs**: dedicated `opencode` section in `docs/Provider_Aliases_User_Guide.md` (full free models table, setup, usage, live-verified notes, stealth model explanation).

### Changed

- `scripts/providers/key-aliases.json` and `scripts/providers/overrides.json` extended with the `opencode` entries (config-only; no code changes — same dynamic pattern as Xiaomi v1.3.0 / Z.AI v1.2.0 / DeepSeek).

### Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 21 new assertions for `opencode`.

### Honest notes

- The alias uses router transport (ccr) because Zen's free models use OpenAI-compatible format (`/v1/chat/completions`), not Anthropic native format. This adds a ccr dependency that native-transport aliases (deepseek, xiaomi) don't have.
- Big Pickle is a stealth model — the actual model served may vary (observed as deepseek-v4-flash). This is by design per OpenCode's documentation.
- The same pre-existing `~/api_keys.sh` set -u issue affects the in-process verifier for all providers; authoritative proof is the direct HTTP round trip.
- The 2 pre-existing, environmental opencode-skill-discovery failures in `run-proof.sh` remain unchanged (unrelated to this work).

## v1.3.0 — 2026-06-19 — Xiaomi MiMo provider alias

### Added

- **`xiaomi` provider alias** — Xiaomi MiMo via the **Anthropic-native endpoint** `https://api.xiaomimimo.com/anthropic` (`POST /anthropic/v1/messages`). Unlike most providers in this toolkit, MiMo exposes a genuine native Anthropic endpoint that accepts `Authorization: Bearer`, so the alias uses **native transport** with no `claude-code-router` (`ccr`) dependency — the same direct-launch model as `deepseek`.

- **Model overrides**: strong = `mimo-v2.5-pro` (flagship, 1M context, reasoning, tool-call), fast = `mimo-v2-flash` (256K, cheapest tier). Pinning is deliberate — models.dev lists a `mimo-v2.5-pro-ultraspeed` id the **live API does not serve**, so the override guarantees only live-served ids are used.
- **key-aliases.json mapping**: `XIAOMI_MIMO_API_KEY` → `xiaomi` (the user's key-var name does not match the models.dev provider's documented `XIAOMI_API_KEY` env).
- **overrides.json pin**: native transport, `/anthropic` base_url, `mimo-v2.5-pro` / `mimo-v2-flash`.
- **Sandbox test coverage**: resolver tests (key-alias mapping, override forces native transport, `/anthropic` base_url beats catalog `/v1`, model pinning beats the stale `ultraspeed` entry, stale-id-never-selected guard) + sync e2e tests (env file, alias, config-dir + plugins symlink, account-detection exclusion, idempotency, no-secret-leak). Providers test 60 → 69 assertions.
- **Live endpoint verification**: `GET /v1/models` HTTP 200 (10 models); native `/anthropic/v1/messages` round trip HTTP 200 with correct text for both `mimo-v2.5-pro` and `mimo-v2-flash`; tool calling proven (`finish_reason: tool_calls`
  - `reasoning_content`); streaming confirmed. Evidence in `scripts/tests/proof/60-xiaomi-live.txt` (secret-free).
- **Docs**: dedicated `xiaomi` section in `docs/Provider_Aliases_User_Guide.md` (model table, setup, usage, live-verified notes).

## Changed

- `scripts/providers/key-aliases.json` and `scripts/providers/overrides.json` extended with the `xiaomi` entries (config-only; no code changes — same dynamic pattern as Z.AI v1.2.0 / DeepSeek).

## Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 9 new assertions for `xiaomi`. Live provider verifier 5/5 PASS.

## Honest notes

- The only failures in the repo's `run-proof.sh` are 2 **pre-existing, environmental** opencode-skill-discovery checks, unrelated to Xiaomi (zero opencode files changed by this release; they fail identically when run standalone).
- The in-process LLMsVerifier step reports `(unverified)` for every provider because `~/api_keys.sh` has a pre-existing unrelated `unbound variable` on a different provider's key under `set -u`;

authoritative proof is the direct native-endpoint round trip (HTTP 200), recorded in the evidence file.

# v1.2.0 — 2026-06-19 — Z.AI Coding Plan provider alias

## Added

- **`zai-coding-plan` provider alias** — OpenAI-compatible router transport via `https://api.z.ai/api/coding/paas/v4` (Coding Max-Yearly Plan endpoint).
- **Model overrides**: strong = `glm-5.2` (flagship 1M context reasoning model, free on plan), fast = `glm-4.7` (204k context, tool_call, 0 cost).
- **key-aliases.json mapping**: `ZAI_API_KEY` → `zai-coding-plan` (targets the coding plan API endpoint instead of the general `z.ai` paas endpoint).
- **overrides.json pin**: overrides auto-selected strong/fast models for the coding plan.
- **Sandbox test coverage**: resolver tests (env-key matching, coding endpoint, router transport, glm-5.2/glm-4.7 model selection) + sync e2e tests (env file, alias, model overrides).
- **Live endpoint verification**: HTTP 200 at `/models` (8 models discovered), curl test of `glm-4.7` chat completion confirmed operational.
- **ccr integration**: provider auto-registered in `~/.claude-code-router/config.json` as the active default route.

## Changed

- `overrides.json` extended with `zai-coding-plan` section for model pinning.

## Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Provider tests include 5 new assertions for `zai-coding-plan`.

# v1.1.0 — 2026-06-16 — Distributed infrastructure + provider verification

Headline: stand up the full LLMsVerifier System on a remote host for heavy testing against **real production LLM services**, plus end-to-end provider aliases proven on two hosts and two transports.

## Added

- **`containers` + `challenges` submodules** (`submodules/`) — the distributed-boot orchestrator and its sibling. `helix-deps.yaml` confirms `containers` has zero own-org submodule deps.
- **Remote host registration** — `config/containers/nezha.env` registers `nezha.local` as a remote boot/test host (SSH key, podman runtime).
- **LLMsVerifier deployment overlays** (`config/containers/llmsverifier/`):
    - `docker-compose.app.yml` — the `llm-verifier` API (cgo image, config mount, `/api/health` healthcheck, loopback, fail-fast secrets).
    - `docker-compose.infra.yml` — observability tier: prometheus + grafana (auto-provisioned datasource + dashboard) + node-exporter. **No DBs** (the app uses SQLite; postgres/redis were unused and removed).
    - `Dockerfile.nezha` / `Dockerfile.mv` — cgo nested-module builds for the server + the `model-verification` tool.
    - `patches/0001..0005` — upstream LLMsVerifier fixes (see PR #2 below).
- **Deployment guide** `config/containers/llmsverifier/README.md` and the **Provider Aliases User Guide** `docs/Provider_Aliases_User_Guide.md` (HTML/PDF/DOCX exports included).
- **QA evidence** `docs/qa/20260616-infra/` — verification proofs, endpoint coverage, security posture, observability, per-provider sweeps, dual-host end-to-end alias proofs.

## Changed

- **Provider session accent color: orange → purple** across spec, guide, and the long-form doc. (Claude Code 2.1.178 cannot persist a default `/color`, so this is the documented default + a manual `/color purple` — a platform limit.)
- `claude-add-account` consolidated onto the shared `cma_link_shared_items` helper (single `CMA_SHARED_ITEMS` source).
- `claude-export-docs` now also emits **DOCX** (HTML/PDF/DOCX).

## Fixed (LLMsVerifier — shipped as PR #2, applied to deployed builds)

- **Auth header missing** — verification requests sent no `Authorization` header → HTTP 401 for every provider. Now `Bearer <key>`.
- **cohere 405** — switched to the OpenAI-compat endpoint (`api.cohere.ai/compatibility/v1`). Verifies at score 1.00.
- **gemini / huggingface** — corrected to OpenAI-compat / router endpoints (huggingface verifies; gemini code-ready pending a valid key).

- **model-id strictness** — verifies a requested id directly when not in the discovered list (no premature `model_not_found`).
- **no `/metrics`** — added `GET /api/metrics` + `/metrics` (stdlib Prometheus).
- **provider-session sync-state noise** — `cma_run_provider` no longer runs cross-account sync-state on isolated provider dirs.

### Verified live (real "Do you see my code?" against production APIs)

- **9 providers verified:** DeepSeek, Groq, Mistral, Cerebras, Novita, NVIDIA, Cohere, Codestral, HuggingFace.
- **Both transports, both hosts:** native (DeepSeek) + router (Novita via ccr) on macOS and on nezha.
- Account-side failures (402/401/429/403) and non-OpenAI providers documented honestly; excluded under "valid users only" but kept fully supported.

### Safety

- Provider dirs (`~/.claude-prov-*`) excluded from account detection — existing `claudeN` accounts and `claude-add-account` untouched.
- Secrets only in the keys file + on-host mode-600 `.env`; never in the repo. All published ports bound to loopback.

## v1.0.0 — 2026-06-16 — Dynamic provider-alias generator

First tagged release. `claude-providers` creates per-provider Claude Code aliases (DeepSeek, Groq, GLM, …) from your keys file pointed at each provider's strongest model — fully dynamic via models.dev + the LLMsVerifier submodule, hybrid native/claude-code-router transport, full lifecycle + tests + docs. See `docs/Provider_Aliases_User_Guide.md`.

## v1.6.7 — 2026-06-21 — Poe proxy fix for all aliases

### Fixed

- **Poe proxy not starting for poe2/poe3** — proxy detection used exact provider ID (`poe2_proxy.py`) which doesn't exist. Fixed to check base name too (`poe_proxy.py` for poe2, poe3 aliases).
- **lib.sh**: base proxy detection with `${CMA_PROVIDER_ID%%[0-9]*}`

- **alias file**: same fix applied

### Verified

- All 3 Poe aliases work: `poe` ✅, `poe2` ✅, `poe3` ✅
- Deployed to both local host and nezha.local

### Tests

- Local: 8/8 ALL GREEN
- nezha.local: 7/8 (pandoc missing — pre-existing)

## v1.6.8 — 2026-06-21 — Poe proxy gzip fix

### Fixed

- **Poe proxy gzip decompression** — Poe API returns gzip-compressed responses but the proxy tried to read them as UTF-8 without decompressing, causing UnicodeDecodeError. Added gzip decompression for both success and error responses.

### Verified

- `poe` (claude-sonnet-4.6): ✅ YES
- `poe2` (gpt-5.5): ✅ YES
- `poe3` (grok-4): ❌ Different error (Grok-4 schema validation, not tools format)

### Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

## v1.6.9 — 2026-06-21 — Poe proxy $ref fix for Grok-4

### Fixed

- **Poe proxy $ref resolution** — Claude Code sends tool schemas with `$ref` references to `$defs`. Grok-4 and some providers don't support `$ref` in tool schemas. Added `resolve_refs()` function that extracts `$defs`, resolves all `$ref` references to inline definitions, and removes `$defs`.

### Verified

- `poe` (claude-sonnet-4.6): ✅ YES
- `poe2` (gpt-5.5): ✅ YES
- `poe3` (grok-4): ✅ YES (was failing, now works)

### Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

## v1.7.0 — 2026-06-22 — Poe proxy complete fix (all aliases verified)

### Fixed

- **Poe proxy shared directory** — the proxy at `~/.local/share/.../proxy/poe_proxy.py` was the OLD version without gzip and $ref fixes. install.sh copies from scripts/ but the shared dir still had the old version. Fixed by ensuring updated proxy is copied to shared directory.
- **install.sh** now copies proxy scripts during installation (already in place)

### Verified (all three aliases through full Claude Code flow)

- `poe` (claude-sonnet-4.6): ✅ YES
- `poe2` (gpt-5.5): ✅ YES
- `poe3` (grok-4): ✅ YES

### Root Cause Analysis

The proxy had three issues:

1. **gzip** — Poe returns gzip-compressed responses, proxy didn't decompress
2. **$ref** — Claude Code sends tool schemas with $ref, Grok-4 doesn't support them
3. **shared dir** — Updated proxy wasn't copied to shared directory

All three fixed and verified.

### Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

## v1.7.1 — 2026-06-22 — Full validation + release

### Fixed

- **Port-ready check** for proxy startup — replaced `sleep 1` with polling loop (`lsof -i`) ensuring proxy is listening before ccr config is written
- **Claude alias regression test** — 11 assertions proving claudeN aliases use `cma_run` (no proxy/transformer code), providers use `cma_run_provider`
- **Command injection fix** in verify_aliases_live.sh — replaced `bash -c` subshell with safe indirect expansion

### Tests

- Local: **9/9 ALL GREEN** (new: test_claude.sh — 11 assertions)
- nezha.local: 8/9 (export fails — pandoc missing)

### Release

- v1.7.1 — pushed to github, gitlab, gitflic, gitverse

## v1.7.2 — 2026-06-22 — Claude alias verification, full release

### Added

- **Claude alias verification** in verify_aliases_live.sh — tests claude1/2/3 alongside provider aliases
- **TOON tested** on all aliases — verified working

### Tests

- Local: **9/9 ALL GREEN**
- nezha.local: 8/9 (pandoc missing)
- All claude1/2/3: ✅ OK
- All provider aliases: verified